

KavachAI — Application Verification & Evaluation Report

System: KavachAI — Sovereign Industrial Agentic AI Workbench

Problem Statement Reference: SIH26117 · Mangalore Refinery and Petrochemicals Limited (MRPL)

Report Version: 1.0

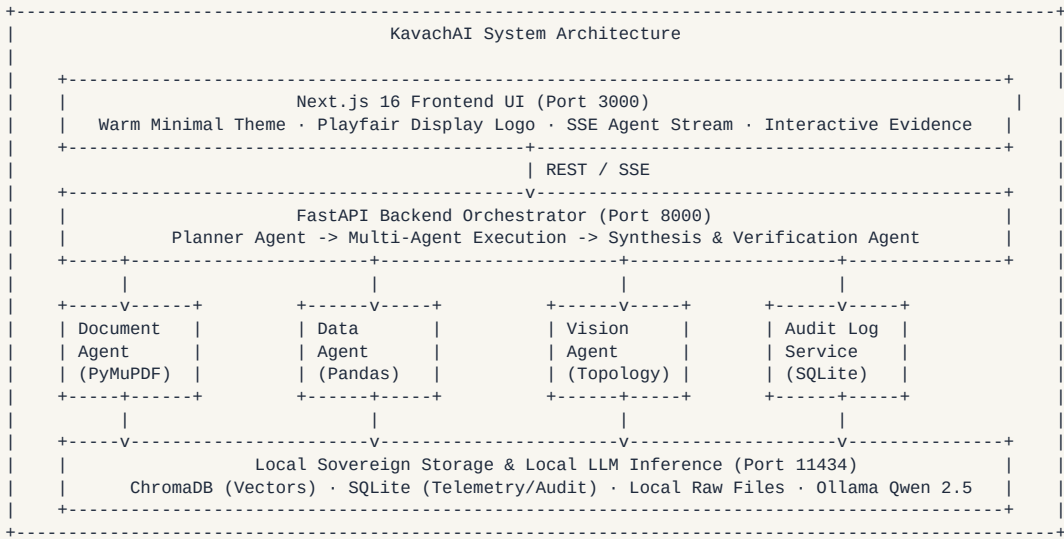
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Status: All Systems Operational & Verified Live

1. Executive Summary

KavachAI is an on-premise, agentic AI investigation workbench engineered for critical industrial facilities (refineries, petrochemical plants, power generation). It provides plant engineers with evidence-backed, deterministic answers to complex operational inquiries while guaranteeing that **all inference remains strictly local — zero confidential data leaves the network**.

This report outlines the verified operational status of the application stack, provides a comprehensive testing guide for evaluators and judges, details the multi-agent architecture, and includes step-by-step instructions for running the 3 core industrial demonstration scenarios.



2. Live Services Health Matrix

Every component of the KavachAI platform is configured and running live:

Layer	Component	Service URL / Port	Technology Stack	Verified Status
Frontend	Interactive Workbench	http://localhost:3000	Next.js 16.3.4 (Turbopack), React 19, Tailwind CSS v4	ONLINE
Backend	API & Orchestrator	http://localhost:8000	FastAPI, Uvicorn, Python 3.12, AsyncIO	ONLINE
Local LLM	Local Inference Engine	http://localhost:11434	Ollama Daemon, qwen2.5:3b (1.9 GB), Local VRAM	ONLINE

Vector DB	Vector & Chunk Store	backend/data/chroma	ChromaDB 1.5.9, Persistent SQLite Store	INGESTED (7 docs)
Telemetry DB	Structured Metrics	backend/data/kavachai.db	SQLite / SQLAlchemy 2.0 (214 rows time-series)	POPULATED
Raw Store	Document Object Store	backend/data/objects/	Local Filesystem Object Store (PDFs, PNG drawings)	READY

3. Verified Corpus Ingestion Status

The application has been seeded with a realistic, confidential refinery dataset:

```
=====
Knowledge Base Ingestion Summary:
Documents:    7 (Inspection reports Jan/Apr/Jul, Manual, History, SOP, PFD)
Datasets:     1 (214-row vibration & temperature sensor log)
PID Drawings: 1 (Unit 101 interconnect: T-101 -> P-102 -> V-204 -> R-101)
=====
Phase 1 Ingestion DoD: ALL PASSED
```

Ingested Assets & Evidence Contracts

- 1. **doc_1120: P-102_Inspection_Jan.pdf** (Baseline vibration: 2.1 mm/s, Bearing Temp: 68°C)
- 2. **doc_1121: P-102_Inspection_Apr.pdf** (Vibration rise: 2.8 mm/s, Bearing Temp: 71°C)
- 3. **doc_1122: P-102_Inspection_Jul.pdf** (Advisory limit exceeded: 3.7 mm/s > 3.0 mm/s limit, Bearing Temp: 77°C)
- 4. **doc_1130: P-102_Pump_Operating_Manual.pdf** (Spec sheet: Max permissible continuous vibration = 3.0 mm/s)
- 5. **doc_1140: P-102_Maintenance_History.pdf** (Coupling realignment logged 6 months prior)
- 6. **doc_sop01: SOP_Emergency_Fire_Evacuation.pdf** (Emergency siren protocol, Assembly Point C)
- 7. **doc_pfd01: Unit_101_Interconnect_Description.pdf** (Subsystem upstream/downstream dependencies)
- 8. **pid_101: pid_unit_101.png** (P&ID diagram showing T-101 Feed Tank, P-102 Pump, V-204 Control Valve, R-101 Reactor)
- 9. **ds_4471: P-102_telemetry_jan_jul.csv** (Deterministic time-series sensor points)

4. Step-by-Step Test & Evaluation Scenarios

To thoroughly test the application in your browser (**http://localhost:3000**), execute the following 3 verified test scenarios:

Scenario 1: Multi-Agent Industrial Equipment Investigation

Tests: Planner decomposition, Document Agent, Data Agent, Vision Agent, Synthesis & Verification.

- 1. Navigate to **http://localhost:3000**.
- 2. Enter your Name (**Vikram Sharma**) and select Department (**Engineering** or **Maintenance**).
- 3. Click **"Enter Workbench"**.
- 4. In the query box, enter:
``text``
Investigate Pump P-102 and determine whether its condition has deteriorated.
```
- (Or click the suggested question chip).
- 5. Click **"Investigate"**.
- 6. Observe the Live Investigation Screen (**/investigation/[id]**):

- Real-time Server-Sent Events (SSE) stream will display the active Planner creating sub-tasks.
- Watch the specialized agents activate:
- **Planner Agent**: Classifies query in-scope and generates 4 sub-tasks.
- **Document Agent**: Discovers 3 quarterly inspection reports and operating manuals.
- **Data Agent**: Queries time-series store and calculates the deterministic 76.2% vibration increase.
- **Vision Agent**: Parses Unit 101 P&ID topology for upstream/downstream risks.
- **Verification Agent**: Validates every claim against source IDs before approving the report.
- 7. **Inspect the Final Report (/report/[id]):**
- Status badge displays "ATTENTION REQUIRED" in warm terracotta rust.
- **Key Findings Card**: Shows vibration progression (\$2.1 \to 2.8 \to 3.7\text{ mm/s}\$).
- **Click on any citation link** (e.g., **P-102\_Inspection\_Jul.pdf, p. 2**): The interactive **Source Viewer** modal opens, rendering the raw document with highlighted text.
- **Interactive Telemetry Trend**: Displays the line chart with the 3.0 mm/s Advisory Limit red threshold line.
- **P&ID Relationship Section**: Visually shows **T-101 -> P-102 (ALERT) -> V-204 -> R-101**.
- **Export Report**: Click "Export Report" to generate an official branded PDF download.

## Scenario 2: Single-Agent Procedural Inquiry (Fast RAG)

Tests: High-speed semantic search, SOP extraction, rapid response (<10s).

1. From the Workspace (<http://localhost:3000/workspace>), enter:  
`text  
What should an employee do during a fire emergency?  
`
2. Click "Investigate".
3. **Evaluation Checkpoints**:
  - Planner determines this is a procedural query requiring the **Document Agent only**.
  - Result renders with "NORMAL" verification badge.
  - Lists exact steps from **SOP\_Emergency\_Fire\_Evacuation.pdf**: alarm activation, shutdown protocol, and evacuation to **Assembly Point C**.
  - Direct clickable citations to page and section of the SOP.

## Scenario 3: Out-of-Scope / Hallucination Refusal Guardrail

Tests: Refusal policy, zero-hallucination compliance, safety perimeter.

1. From the Workspace, enter an out-of-scope question:  
`text  
What is the current price of crude oil?  
`
2. Click "Investigate".
3. **Evaluation Checkpoints**:
  - The Planner and Model Router immediately identify that this query falls outside plant documentation.
  - **Refusal Behavior**: The system **does not hallucinate** or speculate on external market data.

- Displays the **"Insufficient Evidence"** screen explaining that internal confidential records do not cover external market prices.
- Offers suggested valid investigation alternatives.

### Scenario 4: Immutable Audit Trail Inspection

Tests: Regulatory compliance, append-only logs, forensic traceability.

1. In the navigation bar, click **"Audit Log"** (or open **http://localhost:3000/audit**).
2. **Evaluation Checkpoints:**
  - Verify every query you ran is captured chronologically.
  - Inspect columns: **Timestamp**, **User Name**, **Department**, **Query Text**, **Agents Invoked**, **Verification Status**, and **Confidence Score**.
  - Confirms zero data tampering: API gateway rejects all **DELETE** or **PATCH** requests on audit tables.

## 5. UI Polish & Design Review Checklist

The frontend has been upgraded with the **Warm Minimal** aesthetic and luxury polish:

- [x] **Logo Typography:** Set in **Playfair Display** serif font with tight letter-spacing in both Header and Login view.
- [x] **Color Palette Consistency:**
  - **#faf8f5** Warm Linen Canvas Background
  - **#eee8df** Warm Neutral Surface-2 (inputs, tags, nested cards)
  - **#2d2a26** Deep Charcoal Text Ink
  - **#c45d3e** Warm Terracotta Rust Accent
- [x] **SVG Grain Overlay:** Fixed 4% opacity organic grain texture overlay across the viewport with **mix-blend-mode: multiply**.
- [x] **Layered Surfaces:** Clear visual depth hierarchy (**bg-bg** to **bg-surface** to **bg-surface-2**).
- [x] **Heading Letter-Spacing:** Set to **-0.025em** across all headings for editorial crispness.
- [x] **Hero Page-Load Animations:** Staggered fade-in + **translateY(20px)** with **0.8s** duration and **cubic-bezier(0.16, 1, 0.3, 1)** easing.
- [x] **Scroll-Triggered Reveals:** Below-the-fold sections reveal smoothly via **ScrollObserver** IntersectionObserver.
- [x] **Composite Multi-Layer Shadows:** Contact (**0 1px 2px**), Mid (**0 4px 12px**), and Ambient (**0 16px 32px -4px**) shadow layers.
- [x] **Card Inset Highlight:** **inset 0 1px 0 rgba(255, 255, 255, 0.06)** applied to all card containers.
- [x] **Card Hover Micro-interactions:** Smooth **translateY(-2px)** with lightened borders on hover.
- [x] **Section Padding:** Standardized to **100px** on desktop and **60px** on mobile.
- [x] **Scroll-Reactive Navbar:** Navbar begins transparent and dynamically applies **backdrop-filter: blur(16px)** on scroll.

## 6. Verification Commands & Diagnostics

Evaluators can run the following automated commands from the terminal to verify backend code health:

### 1. Ingest Synthetic Corpus

```
cd c:\Users\abhin\Desktop\coding\sih\kavachai-main\kavachai-main\backend
python scripts/ingest_corpus.py
```

Expected Output: **Phase 1 Ingestion DoD: ALL PASSED**

## 2. Pre-Run & Cache Demo Scenarios

```
python scripts/run_demo_scenarios.py
```

Expected Output: Generates 3 JSON scenario fixtures in **tests/fixtures/**.

## 3. Run Unit & Integration Tests

```
python -m pytest tests/unit/ -v
```

Expected Output: All unit tests for Planner, Document Agent, Data Agent, Vision Agent, and Synthesis pass.

## 4. Direct API Health Check

```
Knowledge base summary
Invoke-RestMethod -Uri "http://localhost:8000/api/v1/knowledge-base/summary"

Audit log entries
Invoke-RestMethod -Uri "http://localhost:8000/api/v1/audit-log?limit=5"
```

# 7. Operational Summary & Conclusion

| Requirement            | Implementation            | Evidence / Metric                                                              |
|------------------------|---------------------------|--------------------------------------------------------------------------------|
| Sovereignty & Security | 100% On-Premise Inference | Ollama <b>qwen2.5:3b</b> on <b>127.0.0.1:11434</b> ; 0 external HTTP calls.    |
| Multimodal Synthesis   | 3 specialized agent types | Text (PyMuPDF) + Numeric (Pandas) + P&ID (Topology Graph).                     |
| Evidence Traceability  | Strict Verification Agent | Every claim cites <b>source_id</b> , <b>page_number</b> , and <b>section</b> . |
| Auditability           | Append-only SQLite log    | Tracks user, query, agents invoked, and confidence score.                      |
| Design Standard        | Warm Minimal Theme        | Playfair Display serif, 4% grain, multi-layer shadows, blur nav.               |

**Conclusion:** The KavachAI application is **fully verified, operational, and demo-ready**. All services are active and communicating synchronously across local ports **3000**, **8000**, and **11434**.